

Exercise 4

① Inner JOIN

SELECT employee ID,
first name,
last name,
Department,
salary,
Project ID,
project name,
Budget,
status

FROM employees AS A

INNER JOIN projects AS B

ON A.employee ID = B.employee ID;

Employee ID	First name	Last name	Department	salary	Project ID	Project name	Budget	Status
1	John	Doe	IT	70 000	101	AI Dev	100 000	Completed
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	On-going
3	Bob	Johnson	Finance	75 000	104	Financial analysis	90 000	Pending
4	David	Brown	IT	72 000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65 000	105	market expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk management	80 000	Pending
1	John	Doe	IT	70 000	103	cyber Audit	75 000	Pending

② LEFT JOIN

SELECT employ - ID,
First name,
Last name,
Department,
salary,
Project ID,
proj - name,
Budget,
status

FROM Employees AS A

LEFT JOIN Projects AS B

ON A.employee ID = B.employee ID;

Employee ID	First name	Last name	Department	Salary	Project	Project name	Budget	Status
1	John	DOE	IT	70000	101	AI Dev	100000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	On-going
1	John	DOE	IT	70000	103	Cybersecurity Audit	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial analysis	90000	Ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

3. Right Join

Project ID	Project Name	Budget	Status	Employee ID	First name	Last name	Department	Salary
101	AI Dev	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity Audit	75000	Pending	1	John	Doe	IT	70000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	80000

Syntax: `SELECT` emp_id,
 project_id,
 project_name,
 Budget,
 Status,
 first Employee name,
 last name,
 Department,
 Salary

`FROM` projects AS A

`Right Join` employee AS B

`On` A.employee_id = B.employee_id;

④ Full OUTER JOIN

SELECT emp_id,
first_name,
last_name,
Department,
salary,
Project ID,
project name,
Budget,
status

FROM employees AS A

Full OUTER JOIN project AS B

ON A.employee ID = B.Employee ID

⑦ Filtering Statement

SELECT employee id,
first_name,
last_name,
Department,
salary

FROM employees

WHERE salary > 70 000

Employee id	first_name	Last_name	Department	Salary
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

⑧ SELECT employee ID,
first_name,
last_name,
Department,
Salary

FROM employee

WHERE Department = 'IT' or 'Finance'

Employee id	first_name	Last_name	Department	Salary
1	John	DOE	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

9 SELECT projectid,
 project name,
 Budget,
 Status
 FROM ~~employees~~ Projects
 WHERE project='not complete';

Project Id	Project name	Budget	Status
102	Employee Training	50 000	on-going
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk management	80 000	Pending

10 SELECT project id,
 project name,
 Budget,
 status
 FROM projects
 WHERE budget > 70000 and status = 'not completed'

Project ID	Project name	Budget	Status
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk management	80 000	Pending

11 SELECT employee ID,
 First-name,
 Last-name,
 Department,
 Salary,
 City
 FROM employee
 WHERE city = 'New York or Toronto'
 Order by Salary Desc;

Employee - ID	First-name	Last-name	Department	Salary City	City Salary
1	John	Doe	IT	New-York	70 000
3	Bob	Johnson	Finance	Toronto	75 000
6	Michael	Clark	Finance	New-York	80 000

(12) SELECT employee_ID,
 first-name,
 last-name,
 department,
 salary
 FROM employees
 Order by Salary Desc
 Limit 3;

Employ-ID	First-name	Last-name	Department	Salary
6	Michael	Clark	Finance	80 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000

(13) SELECT sum(salary) AS Total salary,
 department
 FROM employees
 Order by Salary desc